# Memory State: Inference-Native Retrieval for Long-Term Agent Memory

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Code and results: https://github.com/Seetie-AI/memory_state

**Abstract**

Prior work has already shown that prompts can elicit useful text embeddings from generative language-model hidden states. We study a different question: whether those representations are useful as persistent retrieval keys for long-term conversational and preference memory. Memory State captures a suffix-final hidden state from the reasoning model itself, using small memory-specific prompt facets and no trained, distilled, or adapted retriever. The contribution is this agent-memory use and evaluation, not the underlying prompting mechanism.

We construct two retrieval tasks from LongMemEval-S and PrefEval rather than reproduce either benchmark's official end-to-end protocol. On a 100-instance LongMemEval-S-derived turn-retrieval subset, a compact two-view Qwen3.5-9B-MLX-4bit configuration reaches strict Recall@5 of 0.777 using hidden states alone, at 8 KB per page, against 0.755 for a Qwen3-Embedding-8B-4bit-DWQ baseline on the same 94 scored questions. On a PrefEval-derived task that retrieves each question's associated preference from a global pool of 1,000 preference statements, a three-facet hidden-state configuration reaches 0.299 against 0.281 for the same embedding baseline. The results are parity-level evidence that the reasoning model's own memory state can approach a dedicated embedding model on these derived retrieval tasks. A matched 2B-to-9B comparison further moves Recall@5 from 0.596 to 0.691.

## 1 Introduction

File-based agent memory usually begins with plain text and keyword search. It is cheap, transparent, and easy to inspect. It also has a blind spot: the agent must know what to search for. A memory that matters to the current situation may share none of the words in the current query. This is the unknown unknown problem.

Semantic retrieval addresses it, but it adds a second model. That retriever is usually smaller than the reasoning model, and it was trained for a retrieval objective that may not match the agent-memory domain it is deployed in. If it misses, the more capable model never sees the memory.

This work tests a third arrangement. The model that reads and reasons over a memory also produces the vector that finds it later. If that works, the memory index shares an inference path, a tokenizer, and a semantic space with downstream reasoning, and it needs no separately trained component.

The question came from an anomaly. While sweeping prompt suffixes on the PrefEval-derived retrieval task, a generic prompt asking the model for a single emoji reached Recall@5 of about 0.285, next to 0.281 for an 8B embedding model on the same 1,000 examples. That prompt was not the best in the sweep and it is not part of any system reported here. We record it as the observation that motivated the work: a reasoning model appears to carry a usable memory representation in places we do not normally treat as an index.

We report three things. First, hidden-state keys reach embedding-baseline scale on two benchmark-derived memory retrieval tasks. Second, three simple memory facets selected on conversational memory remain useful on preference retrieval. Third, the representation improves with model size under matched conditions.

## 2 Related Work

**PromptEOL** [1] adds an explicit one-word constraint to a causal language-model prompt and uses the hidden state at the suffix-final input position, which predicts the next token, as a sentence embedding without fine-tuning.

**MetaEOL** [2] extracts multiple one-word-limited views for meta-tasks such as classification and sentiment, then averages their embeddings. It also reports that the final layer is not always best and that a proportional layer-selection strategy improves its largest-model result.

**PromptReps** [3] applies the one-word prompt to training-free full-corpus document retrieval. It combines the last layer's suffix-final hidden state with logits for the next token to form a dense and sparse hybrid, processes queries with a corresponding query prompt, and reports stronger retrieval with a larger LLM.

**All-but-the-Top** [4] showed that subtracting the common mean and projecting word vectors away from a small number of dominant principal directions can improve their quality across several tasks.

Memory State does not claim these mechanisms. Prompted suffix-final hidden states, multiple prompt views, layer selection below the output layer, and dominant-direction removal all precede this work. We combine them for a different use: persistent retrieval keys inside an agent's own inference path, with small prompt facets aimed at user, conversation, and association memory.

**LongMemEval** [5] evaluates final question answering over long multi-session histories and provides answer-location annotations for intermediate retrieval analysis. **PrefEval** [6] evaluates preference following through long-form generation or four-option classification; its retrieval-augmented setting retrieves conversation exchanges before answer generation. Our experiments below derive retrieval-only tasks from their data. They are not official LongMemEval or PrefEval leaderboard evaluations.

# 3 Method

For each memory page and each query, the model is run over the text plus a retrieval suffix that asks it to compress the text into a short key. The retrieval vector is the hidden state at the final token of that suffix.

The pattern is:

1. choose prompt views that match the memory domain;
2. extract the suffix-final hidden state from a late-layer band near the output;
3. apply dominant-direction removal to strip shared prompt and corpus geometry;
4. combine views by concatenation or component-normalized vector averaging;
5. optionally add a lexical score as a shortlist rerank signal.

Queries are encoded with the same recipe as memory pages. We tested the asymmetric alternative, encoding memory and query with different prompts, and it did not help: the best asymmetric configuration reached Recall@5 of 0.755 against 0.766 for symmetric encoding of the same prompt.

The reported configurations use hidden states as the dense representation. Next-token logits were also collected, and a PromptReps-style sparse path was reproduced, but it is not part of the reported systems.

## 3.1 Prompt views

Suffixes are written in Chinese, ask for one word, and decode one token. The three semantic facets in the transferred K3 configuration are, translated:

```
Use one word to represent the user in the conversation above. The word is: "
Use one word to mark the conversation above. The word is: "
Use one word to represent the association this conversation creates for me.
The word is: "
```

The trailing quotation mark is part of the suffix. It opens the model's answer, so the next decoded token is the key itself. These are deliberately small semantic tweaks, not new prompting machinery. When the facets were evaluated on the PrefEval-derived task, the Chinese verb for "represent" in the user and association views was changed to "mark" to match that sweep's prompt family. Thus the transferred unit is the three semantic facets, not byte-identical prompt text. The exact Chinese suffixes and full configurations are in the repository.

## 3.2 Geometry

Dominant-direction removal is applied symmetrically to query and candidate vectors with rank 15. Across 17 symmetric prompt variants, symmetric removal won Recall@5 for 13, query-only removal for 3, and plain centered cosine for 1.

# 4 Experimental Setup

Hidden-state extraction uses `Qwen3.5-9B-MLX-4bit`, with `Qwen3.5-2B-MLX-4bit` for the model-size check. Embedding baselines are `Qwen3-Embedding-8B-4bit-DWQ` and `Qwen3-Embedding-0.6B-4bit-DWQ`, from the Qwen3 Embedding series [7]. All models are local 4-bit conversions. Embedding scores are never concatenated into the hidden vectors; they appear only as baselines or as score-level signals.

We do not report either benchmark's official end-to-end score. For the LongMemEval-S-derived task, each question ranks only the user turns from its own haystack sessions; candidates from different questions are never mixed. We evaluate a 100-instance subset, of which 94 questions remain after abstention and missing-user-label filtering. Recall@5 is stricter than a first-hit measure: a question scores 1 only if every gold user turn appears in the top 5. We do not run the benchmark's final answer-generation stage.

For the PrefEval-derived task, we take 1,000 implicit-persona rows, place their ground-truth preference statements in one global candidate pool, and ask each row's final question to retrieve its paired preference statement. Unlike LongMemEval, candidates from different rows are intentionally mixed. This does not reproduce PrefEval's full context, long-form generation, or four-option classification. It isolates preference-memory indexing.

Benchmark data is not redistributed with the repository. Sources and download instructions are documented there.

# 5 Results

## 5.1 LongMemEval-derived turn retrieval

Table 1: Retrieval results on a 100-instance LongMemEval-S-derived turn-retrieval subset.

| Method | R@5 | NDCG@5 | MRR |
|---|---|---|---|
| BM25 | 0.585 | 0.547 | 0.559 |
| Qwen3-Embedding-8B-4bit-DWQ | 0.755 | 0.789 | 0.826 |
| Memory State, compact K2, hidden only, 8 KB/page | 0.777 | 0.788 | 0.820 |
| Memory State, K3 concat plus BM25 top 20, 24 KB/page | 0.777 | 0.822 | 0.851 |

*Metric guide, for this and every table below. R@k (Recall@k) measures how much relevant memory evidence appears within the first k results; LongMemEval uses a strict version requiring every gold turn in the top five. NDCG@5 rewards relevant evidence appearing nearer the top. MRR measures how early the first relevant result appears. Higher is better throughout.*

The compact two-view configuration matches the three-view hybrid on strict recall while using no lexical signal and one third of the storage. Its 0.022 margin over the 8B embedding baseline equals two questions in this sample. At that resolution we make no significance claim. This is parity-level evidence at embedding-baseline scale without training an embedding model, not a decisive held-out win.

## 5.2 PrefEval-derived preference retrieval

The three semantic prompt facets were selected on the LongMemEval-derived task. They were then evaluated on the PrefEval-derived global retrieval task over all 1,000 preference statements. Prompt wording followed the PrefEval sweep's "mark" family, and per-view layer and dominant-direction settings were assigned there. What transfers is the user, conversation-tag, and association decomposition, not exact prompt text or a frozen extraction configuration.

Table 2: Retrieval results on a PrefEval-derived global preference-retrieval task.

| Method | R@1 | R@3 | R@5 | NDCG@5 | MRR |
|---|---|---|---|---|---|
| BM25 | 0.035 | 0.074 | 0.094 | 0.066 | 0.067 |
| Qwen3-Embedding-8B-4bit-DWQ | 0.093 | 0.216 | 0.281 | 0.191 | 0.200 |
| Memory State, LongMem prompt transfer, hidden only | 0.093 | 0.212 | 0.299 | 0.197 | 0.188 |

The transferred facet combination exceeds the 8B embedding baseline on Recall@5 using no lexical signal and no embedding score. It is slightly lower on R@3 and MRR. This is a comparison within our derived retrieval task, not a PrefEval benchmark score. The LongMemEval shortlist and fusion weights were not carried over.

Two things make this the most informative PrefEval row we have. It was not selected to win here: of the four K3 candidates evaluated on PrefEval it placed last, below combinations whose prompts were chosen on the PrefEval-derived task itself. And its semantic facets were fixed by a different benchmark-derived task. So it is the weakest candidate in its own sweep and still clears the 8B embedding baseline on strict recall. This supports the claim that a prompt combination can cross memory domains, not that every low-level parameter is universal.

A separate configuration tuned directly on the PrefEval-derived task, fusing the hidden-state score with an embedding score and BM25, reaches Recall@5 of 0.332. Because its prompts and fusion weights were selected on the same data it is reported on, we treat it as a domain-tuned exploratory result and not as the headline.

### 5.3 Model size

Both models were run with the same prompt suffix, the same centered transform, the same 4-bit precision, and the same 100 instances.

Table 3: Matched model-size comparison.

| Model | Layer | R@5 | NDCG@5 |
|---|---|---|---|
| Qwen3.5-2B-MLX-4bit | 22 | 0.596 | 0.547 |
| Qwen3.5-9B-MLX-4bit | 30 | 0.691 | 0.689 |

The larger model gains 9.5 percentage points on Recall@5. On a smaller 30-instance tuning slice both models scored 0.833, so the gap appears only at the larger sample. Each model uses its own best late layer rather than a fixed absolute index. The direction is consistent with earlier retrieval results reported for prompted representations at larger model scale [3].

## 6 Analysis and Negative Results

**There is no universal best layer.** Across the prompt sweep, layers 29, 30, and 31 each won for 6, 6, and 5 variants respectively. The useful region is a late-layer band whose exact position depends on prompt semantics, not a single layer.

**Prompt wording matters more than expected.** Changing one word in an otherwise identical suffix, from a neutral term for the other party to the product-native term for the user, moved Recall@5 by 0.043. The model appears better aligned to product vocabulary than to neutral phrasing.

**Prompt language is not neutral.** In the sweep, the best cell for an English variant of one suffix reached Recall@5 of 0.713 while the best cell for its Chinese original reached 0.755, on English memory content. The two best cells do not share a layer and transform, so this is a sweep observation rather than a single-variable comparison. Cross-language prompting worked here; we make no general multilingual claim.

**Not every retrieval-shaped prompt works.** An answer-strategy prompt fell to Recall@5 of 0.574.

**Lexical fusion helps only as a small vote inside a shortlist.** Fusing BM25 inside the vector top 20 or top 50 beat full-corpus fusion for ranking, and weighting BM25 heavily collapsed performance across configurations, because turn-level lexical matching is noisy.

**Late interaction did not justify its cost.** Max-sim style scoring over multiple stored vectors did not beat the simpler concatenation and averaging candidates.

**Agreement filtering is a diagnostic, not a default.** Gold candidates were much more likely than non-gold candidates to appear in multiple prompt rankings, but sorting by agreement never beat sorting by score.

**A stronger reranker buys ranking, not recall.** Blending an 8B embedding score into the hybrid raised NDCG@5 to 0.834 and MRR to 0.888 without improving Recall@5, while reintroducing the separate model dependency the method exists to avoid. We keep it as a ceiling, not a default.

**Small subsets misled us.** An early prompt that led a 40-instance slice fell to mid-pack on the full 100 instances. Oracle union diagnostics, which use gold labels to find complementary prompt pairs, were likewise treated as hypothesis generators rather than estimates.

# 7 Deployment Considerations

The vector comes from the final token of a prompted key-generation pass. In an online setting the reasoning model has already processed the conversation and its context is in the KV cache, so capturing a memory key means decoding one additional token per prompt view and keeping its hidden state. Memory capture then happens inside the inference path rather than as a second encoding job by another model. We did not measure end-to-end latency or cost, so this is a deployment argument, not a measured result.

Storage is straightforward. Qwen3.5-9B has a hidden size of 4096, so one view in bf16 costs 8 KB per page and three views cost 24 KB. Around 20,000 memory pages need roughly 500 MB before index overhead. The compact configuration in Section 5.1 reaches the same strict recall as the three-view hybrid at 8 KB.

# 8 Limitations

The reported numbers are for retrieval tasks derived from LongMemEval and PrefEval data, not their official end-to-end protocols. In particular, we do not measure LongMemEval answer accuracy or PrefEval generation/classification accuracy. The LongMemEval-derived numbers rest on 94 scored questions, where one question moves the metric by about 1.1 percentage points. Differences of one or two questions between the reported systems and the embedding baselines should be read as parity, not as ordering.

The PrefEval-derived global pool labels only each paired preference statement and 10 exact textual duplicates as gold. In a manual audit of 100 sampled misses, 58 were plausible same-topic candidates, including some semantically equivalent false negatives; 28 shared the topic but carried the wrong constraint, such as an allergy confused with a stylistic dislike; and 14 were unrelated. Recall@5 here therefore measures paired-memory recovery rather than utility. It may understate useful retrieval, but simply crediting same-topic hits would hide real and sometimes safety-critical errors.

Prompt and fusion choices were selected on explored benchmark subsets. The LongMemEval-derived results are therefore exploratory rather than a clean held-out evaluation. The PrefEval-derived result in Section 5.2 is the closest we come to an independent estimate, because its semantic facets were fixed by a different task and it was not the configuration selected to perform best there. Its prompt wording, per-view layer, and geometry settings were still assigned on the PrefEval side, so it is not a held-out evaluation either.

The geometry is tied to `Qwen3.5-9B-MLX-4bit` and may change under another model, another quantization, another language, or another prompt template. Both models tested come from one family, so we cannot say whether the effect holds across model families.

The work supports feasibility at embedding-baseline scale. It does not establish a general win over embedding models.

# 9 Reproducibility

Code, experiment logs, the earlier internal results overview, and the result files for all reported numbers are public at https://github.com/Seetie-AI/memory_state.

Benchmark datasets and model weights are not redistributed. The repository documents the original sources for LongMemEval and PrefEval, the download commands, and the model identifiers, so that users obtain them under the original terms.

# 10 Conclusion

A generative model, in the course of reading a page, leaves behind a representation that can be used to find that page again. On retrieval tasks derived from two different memory benchmarks, this representation performed at the scale of a dedicated embedding model without training a retriever, and it improved with model size under matched conditions.

The claim is not that hidden states beat embedding models. It is that an agent's memory index may not require a second model at all, and that the memory an agent needs may already be present in the act of understanding, waiting to be read out rather than rebuilt.

## References


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